

1. Explain Programming and Python in Detail.

Programming is the process of creating a set of instructions that tell a computer how to perform a task. Python is a popular, general-purpose programming language known for its simple and readable syntax.

→ It is essential for creating software, websites, and allow humans to interact with computers and automate tasks.

Characteristics of Python:

- Simple and Readable: Python's syntax is similar and easy to learn, compared to other languages.
- Interpreted: Code is executed line by line, which speeds up the prototyping process.
- Dynamically Typed: Data types are determined at runtime, offering flexibility.
- Large Standard Library: Provides ready to use tools and third party libraries.
- Supports multiple programming paradigms: It can be used for procedural, object-oriented and functional programming.

Applications of Python:-

- Software development
- Data Science.
- Data Analytics.

→ Machine learning

Types of comments in python:

- Single line Comment: Start with a hash (#) symbol. Anything after the (#) on the same line is ignored by the interpreter.
- Multi line comment: While python doesn't have a specific syntax for multi-line comments like other languages. It can be used {`"""` (or) `'''`} . are often used as documentation strings.

Importance of Python:

- Open-Source and free to use
- Simple Syntax. makes coding faster and efficient
- Growing demand for python developers in job market
- Great for Data Analysis and Visualisation
- Works on many computers and platforms
- Lots of libraries and frame works like Pandas
- Easy to learn and use, great for beginners
- Used in web development, data Science and more.
- It is simplicity and readability make it ideal for rapid development.

2. Describe Data types and Operators in Python with suitable examples

Python includes several built-in data types and a variety of operators to manipulate data. Understanding these is fundamental to programming in python.

Data types:

These data types are used to store various forms of data. You can determine the type of any variable using the built-in `type()` function.

- Numeric Data types : It is used to store numbers

→ int: → complex :-

$$\begin{aligned} x &= 10; & x &= 2+3j \\ x &= -5; \end{aligned}$$

→ float:

$$\theta = 20.5^\circ$$

Ex:- int $\rightarrow$ no of students

float $\rightarrow$ weight, height

- Sequence Data types: It is used to store multiple values in order.

→ string : used for text.

- name = "keertana".

→ List :- Mutable { can change values }

- fruits = ["apple", "Banana", "orange"]

→ Tuple: Immutable { cannot change values }

- `days = ("Mon", "Tue", "Wed")`.

→ set: unique values (no duplicates)

• numbers = {1, 2, 3, 3}

Output: {1, 2, 3}

→ Mapping Data type: Key-Value form

• student = {"name": "keerthana", "age": 21}

→ Boolean Data type: stores true or false

• x = 25

print
(x > 25)

Output: True

Operators in Python: It performs operations on variables and values

(1) Arithmetic Operator: Used for mathematical calculations.

<u>Operator</u>	<u>Use</u>	<u>Example</u>
+	Addition	a = 10 b = 3
-	Subtraction	print(a - b)
*	Multiplication	Output: 13
/	Division	
%	Remainder	

(2) Assignment Operators: It is used to assign and update values.

→ x = 5

→ x += 2

→ x -= 2

→ x *= 35

(3) Comparison Operators :- Compare two values and Return true or False.

operator	Meaning.
<code>==</code>	Equal
<code>!=</code>	Not equal
<code>></code>	Greater than
<code><</code>	less than
<code><=</code>	less than and equal to
<code>>=</code>	greater than and equal to

Ex:- `a=10`

`b=5`

`print (a>b)`

output: ~~TRUE~~

(4) Logical Operators: It is used to combine Conditions.

operator	Use.
and	Both conditions true
or	Any one condition true
not	Reverse Result

Ex: `print (10>5 and 5>2)`

output: True.

Membership Operators: Check whether a value exists in a sequence.

operator	Meaning
in	present
not in	Not present

Ex:- print ("apple" in fruits).

Identity Operators: check whether two variables refer to the same object.

operator	Meaning
is	same object
is not	different object

Ex:- a=10

b=10

print (a is b).

⇒ Python provides various data types to store different kinds of data and operators to perform operations..

(3) Explain Python Input and Output Operations in detail

Python Input and Output operations are used to take data from the user and display result to the user.

→ Input Operation

The input() function is used to accept input from the user

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through the keyboard.

```
Ex: name = input("Enter your name:")  
print(name).
```

By default, `input()` always returns data in string (str) format even if the user enters numbers.

```
Ex:- age = input("Enter age:")  
print(type(age))
```

output: <class 'str'>.

Runtime Input:- since input received as a string, type conversion is needed to perform calculations.

common type conversions:

- `int()` - converts to integer
- `float()` - converts to decimal
- `str()` - converts to string

```
Ex:- age = int(input("Enter your age:"))  
print(age+1).
```

Multiple Inputs:-

We can accept ^{multiple} inputs at a time.

```
Ex:- a = int(input("Enter first number:"))  
b = int(input("Enter second number:"))
```


* Output Operation:-

→ `print()` function is used to display output on the screen.

Ex:- `print("Hello")`.

Output: 'Hello'

* Using Separators (sep)

The `sep` parameter changes the separator between values.

Ex:- `print("python", "Java", "C", sep = ",")`

Output:

python, Java, C.

* Using end parameter :-

It changes what is printed at the end of the line.

Ex:- `print("Hello", end = " ")`

`print("World")`

Output:

Hello World.

* Formatted Outputs-

Formatted output helps display data neatly and clearly.

using comma

`name = "keerthana"`

`age = 21`

`print("Name = ", name, "Age = ", age)`

* Using format() method:-

name = "Raj"

age = 25

print("My name is {} and my age is {}".format(name, age))

* Using Format Specifiers (%)

specifier meaning.

%d Integer

%f float

%s string

Ex:-

age = 21

height = 5.3

print("Age = %d, Height = %f")

→ Input() is used to take input and always returns string

→ Multiple inputs is allowed in python

→ print() is used to display output

→ Formatted Outputs improves readability.

→ Type conversions is needed to perform Calculations.

4) Control Statements and Decision Making Statements in Python:

Control Statements are used to control how a program runs. Normally, a python program runs line by line, but control statements allow us to:

Take decisions

Repeat tasks

Stop or Skip some statements

They make program logical, useful and flexible.

Types of control Statements in Python:

Python control statements are mainly of three types:

(1) Decision Making Statements:

Decision making statements helps a program to choose what to do based on conditions.

i) if Statement:

The if statement runs code only when the condition is true.

Syntax:

if condition:

statement

Ex:-

```
age = 20
```

```
if age >= 18:
```

```
    print("you can vote")
```

Execution:-

If the condition is true $\rightarrow$ code runs
If false $\rightarrow$ code is skipped.

(2) if-else statement:

The if-else statement runs one block if the condition is true, otherwise another block.

Syntax:

if condition:

statement 1

else:

statement 2

Exam:-

```
num = 5
```

```
if num % 2 == 0:
```

```
    print("Even number")
```

```
else:
```

```
    print("odd number")
```

Execution:

(3) if-elif-else statement:

Used when there are many conditions

Syntax:

if condition 1:

statement 1

elif condition 2:

statement 2

else:

statement 3

Example

```
marks = 70
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Grade C")
```

Execution:

- Conditions are checked one by one
- first true condition runs
- if none are true, else runs.

Conclusions:-

Control statements help the control flow of a python program

Decision Making statements help the program take decisions based on conditions and make programs smarter.

Control statements are fundamental constructs that determine the order in which instructions are executed in a program. They are crucial because they enable the program to make decisions and execute specific code.

This allows for the creation of effective programs that can respond differently to various situations.

(5) Write an essay on Python programming fundamentals?

Programming plays an important role in problem solving. It helps us break a problem into smaller steps and give instructions to the computer to solve it. By using programming we can automate tasks, reduce human effort and get accurate results quickly. Programming is used in many fields such as education, business, science and technology. It provides a structured approach to solving problems. It allows a person to translate a problem into a sequence of logical instructions that a computer can execute. This process involves problem analysis, Algorithm, Coding, Testing and Debugging.

Python syntax is designed to be clear, simple and highly readable, making it beginner friendly. It emphasises code readability which means the code is easier to write, understand and maintain. It involves minimalist grammar, indentation, straight forward keywords. This simplicity is a primary reason for its popularity and ease of learning.

Use of comments :

Comments are used to add explanatory notes within the code, improving documentation and clarity for developers.

→ Comments are non-executable lines of text that the python interpreter ignores.

It focuses to explain the user about the purpose of specific line or block of code.

→ It maintains complex logic easier for other programmers to understand.

→ To document the code's functionality and design

→ In python comments are typically created using the hash(#) symbol

Data types, Operators and input/output operations:

→ Python is dynamically typed and supports fundamental data types such as integer (int), floating-point numbers (float), strings (str), booleans (bool).

→ Operators includes arithmetic operators (+, -, *, /), comparison operators (==, >, !=, <) and logical operators (and, or, not) and other operators are used to perform operations on the data.

→ Input/output functions like input() function is used to get data from the user and print() function is used to display output to the console.

Python supports various data types and operators for manipulating data, along with built-in functions for handling input and output.

PROGRAMS

1) Program:

```
age = int(input("Enter age: "))
is3D = int(input("Is it 3D movie? (1=yes, 0=no): "))
price = 0
if age < 13:
    price = 150
elif age <= 59:
    price = 250
else:
    price = 200
if is3D == 1:
    price = price + 50
print("Final Ticket Price:", price)
```

#output:-

Enter age : 35

Is it 3D movie? (1=yes, 0=no): 1

Final Ticket Price : 300.

2) Program:

```
att = int(input("Enter attendance Percentage: "))
med = int(input("Medical Certificate: "))
if att >= 75:
    print("Allowed")
elif att >= 60 and med == 1:
    print("Allowed")
else:
    print("Not Allowed")
```

#output:-

Enter attendance percentage : 65

Medical Certificate : 0

Not Allowed.

(3) Program:- E-commerce Discount

```
bill = int(input("Enter bill :"))
isprime = int(input("prime num :"))
if bill > 50000:
    discount = bill * 20/100
elif bill >= 2000 or bill <= 4999:
    discount = bill * 10/100
else:
    discount = 0
if isprime == 1:
    discount += 0
final_amount = bill - discount
print("Final amount price:", final_amount)
```

output

Enter bill : 70

prime num : 1

Final amount price : 59.5

(4) Program:- Smart Phone Battery Warning.

```
battery = int(input("Enter battery percentage :"))
charging = int(input("Is phone charging? (1=yes, 0=no)"))
if charging == 1:
    print("charging")
else:
    if battery <= 20:
        print("Low Battery")
    elif battery <= 80:
        print("Normal")
    else:
        print("Full")
```

output

Enter battery percentage : 10

Is phone charging? (1=yes, 0=no) : 1

Low Battery

5) Program:- Driving License.

```
age = int(input("Enter age:"))
```

```
Testpassed = int(input("Test Passed : "))
```

```
if age >= 60:
```

```
    print("Eligible")
```

```
elif age >= 18 and Testpassed == 1:
```

```
    print("Not Eligible")
```

```
else :
```

```
    print("Not Eligible")
```

output:

Enter age : 28

Test passed : 1

Eligible!

6) Program:- Online Food Delivery.

```
amt = int(input("Enter Order amt:"))
```

```
isGold = int(input("Gold Member? : "))
```

```
dist = int(input("Enter distance : "))
```

```
if dist > 10:
```

```
    print("Delivery Charged")
```

```
elif amt >= 500 or isGold == 1:
```

```
    print("Free Delivery")
```

```
else :
```

```
    print("Delivery Charged")
```

#output:-

Enter Order amt : 600

Gold Member? : 1

Enter distance : 17

Delivery Charged.

(7) Program: Bank loan Approval.

```
sal = int(input("Enter your salary:"))  
cr = int(input("Enter your credit score:"))
```

```
if sal >= 30000 and cr >= 700:
```

```
    print("Loan approved")
```

```
elif sal >= 50000:
```

```
    print("Loan approved")
```

```
else:
```

```
    print("Loan rejected")
```

output:-

Enter your salary: 35000

Enter your credit score: 900

Loan Approved.

(8) Program: Electricity Bills.

```
units = int(input("Enter Units:"))
```

```
if units <= 100:
```

```
    bill = units * 2
```

```
elif units <= 200:
```

```
    bill = (100 * 2) + (units - 100) * 3
```

```
else:
```

```
    bill = (100 * 2) + (100 * 3) + (units - 200) * 5
```

```
print("Final Bill amount:", bill)
```

output:-

(9) Program: Student Scholarship.

```
marks = int(input("Enter marks: "))
income = int(input("Enter income: "))
singleparent = int(input("Singleparent (1=yes, 0=no): "))
if marks >= 85 and (income < 500000 or singleparent == 1)
    print("Scholarship Approved")
else:
    print("Scholarship Not Approved")
```

#Output:

Enter marks: 90

Enter income: 30000

singleparent: 1

Scholarship Approved.

(10) Program: Online Exam Result.

```
theory = int(input("Enter theory: "))
practical = int(input("Enter practical: "))
```

```
total = theory + practical
```

```
if (theory >= 40 and practical >= 40) or total >= 100:
```

```
    print("pass")
```

```
else:
```

```
    print("fail").
```


11. Program:-

```
days = int(input("Enter Days Stayed:"))
```

```
weekend = int(input("Weekend (1/0):"))
```

```
if weekend == 1 and days > 3:
```

```
    bill = 4000 * days
```

```
    bill = bill - (bill * 0.15)
```

output :

```
elif weekend == 1:
```

```
    bill = 4000 * days
```

```
elif days > 3:
```

```
    bill = 3000 * days
```

```
    bill = bill - (bill * 0.15)
```

```
else :
```

```
    bill = 3000 * days
```

```
print("Total Bill: ", bill)
```

Enter Days Stayed : 4

Weekend (1/0) : 0

Total Bill : 10200

(12) Program:

```
score = int(input("Enter Score:"))
```

```
isPremium = int(input("Premium:"))
```

```
usedCheat = int(input("UsedCheat:"))
```

```
if usedCheat == 1:
```

```
    print("Access Denied")
```

```
elif score >= 100 or isPremium == 1:
```

```
    print("Game Unlocked Next Level")
```

```
else :
```

```
    print("Access Denied")
```

output:

Enter Score : 100

Premium : 0

UsedCheat : 0

Game Unlocked Next Level

(13) Program:

```
dataUsed = float(input("Enter Data Used : "))  
hasUnlimited = int(input("Unlimited Plan : "))  
isRoaming = int(input("Roaming : "))  
if isRoaming == 1:
```

```
    print("Unlimited Data doesn't work")
```

```
elif dataUsed >= 2 or hasUnlimited == 1:
```

```
    print("Unlimited Data")
```

```
else:
```

```
    print("Limited Data")
```

(14) Program:

```
idValid = int(input("ID Valid : "))
```

```
fingerprint = int(input("Fingerprint : "))
```

```
faceScan = int(input("Face scan : "))
```

```
isHoliday = int(input("Holiday : "))
```

```
if isHoliday == 1:
```

```
    print("Entry Denied for everyone")
```

```
elif idValid == 1 and (fingerprint == 1 or faceScan == 1):
```

```
    print("Entry Allowed into Office")
```

```
else:
```

```
    print("Entry Denied")
```

output:

Enter Data Used : 1

Unlimited Plan : 0

Roaming : 1

Unlimited Data doesn't work

output:

ID Valid : 1

Fingerprint : 1

FaceScan : 1

Holiday : 1

Entry Denied for everyone.

(15) Program.

```
avg = float (input ("Enter Average Rating:"))
```

```
editor = int (input ("Editor's Choice:"))
```

```
if editor == 1:
```

```
    print ("Recommended")
```

```
elif avg >= 8.5:
```

```
    print ("Excellent")
```

```
elif avg >= 6.0 and avg <= 8.4:
```

```
    print ("Good")
```

```
else:
```

```
    print ("Average").
```

output:

Enter Average Rating: 5

Editor's Choice: 0

Average.